

Smart Product Intelligence

Final Report

Capstone Project — Khazar University
Hands-on Deep Learning Course | Author: Eltun | 2025

1. Project Overview

This project applies the full spectrum of modern deep learning to a single real-world problem: understanding and assisting buyers of toys on Amazon. A single coherent dataset (Amazon Reviews 2023, Toys & Games category) is attacked with progressively more powerful tools, from a logistic-regression baseline through transformers and diffusion models, then integrated into one working assistant.

Dataset: 15,000 unique products, 75,000 reviews, ~8,000 cached product images. Splits are **product-level** (Train 12,000 / Val 1,500 / Test 1,500) with zero overlap between sets — the brief is explicit about avoiding review-level leakage.

2. Results Summary

Milestone	Task	Baseline	Deep Model	Improvement
M1	Rating band (tabular)	LogReg F1 = 0.372	MLP F1 = 0.428	+15.0%
M2	Image subcategory	CNN scratch (overfits)	MobileNetV2 acc = 0.631	Transfer wins
M3	Sentiment	TF-IDF F1 = 0.864	Embeddings F1 = 0.826	TF-IDF wins (honest)
M4	Sentiment (transformer)	TF-IDF F1 = 0.864	DistilBERT F1 = 0.891	+2.6 pp

3. Key Honest Findings

- TF-IDF beats sentence embeddings on sentiment.** Embeddings are 7× slower and slightly less accurate. Bigger model is not always better — choose representations based on evidence, not on which is newer.
- DistilBERT accuracy drops with review length.** From ~95% on short reviews to ~88% on 500+ character reviews. The 128-token limit and mixed sentiments in long reviews explain this.
- Transfer learning matters in computer vision.** MobileNetV2 reaches ~66% accuracy with half the parameters of a from-scratch CNN, which severely overfits the limited training data.
- RAG reduces LLM hallucination by ~88%.** Ungrounded LLM invented 8 fake product names across 5 questions; grounded RAG invented 1.
- Diffusion fails on branded items.** Funko, Playmobil, licensed characters are not recognized — generation is useful for marketing imagery, not as a product-photo replacement.

4. Milestone 0 — Data Preparation & EDA

Exploratory analysis revealed three issues that motivated later modelling decisions: (1) a heavy 5-star rating imbalance (66% of reviews), which forces macro-F1 reporting; (2) 43% of products have missing prices, handled with a missing-data flag rather than row deletion; (3) image links break over time, hence local caching during M0.

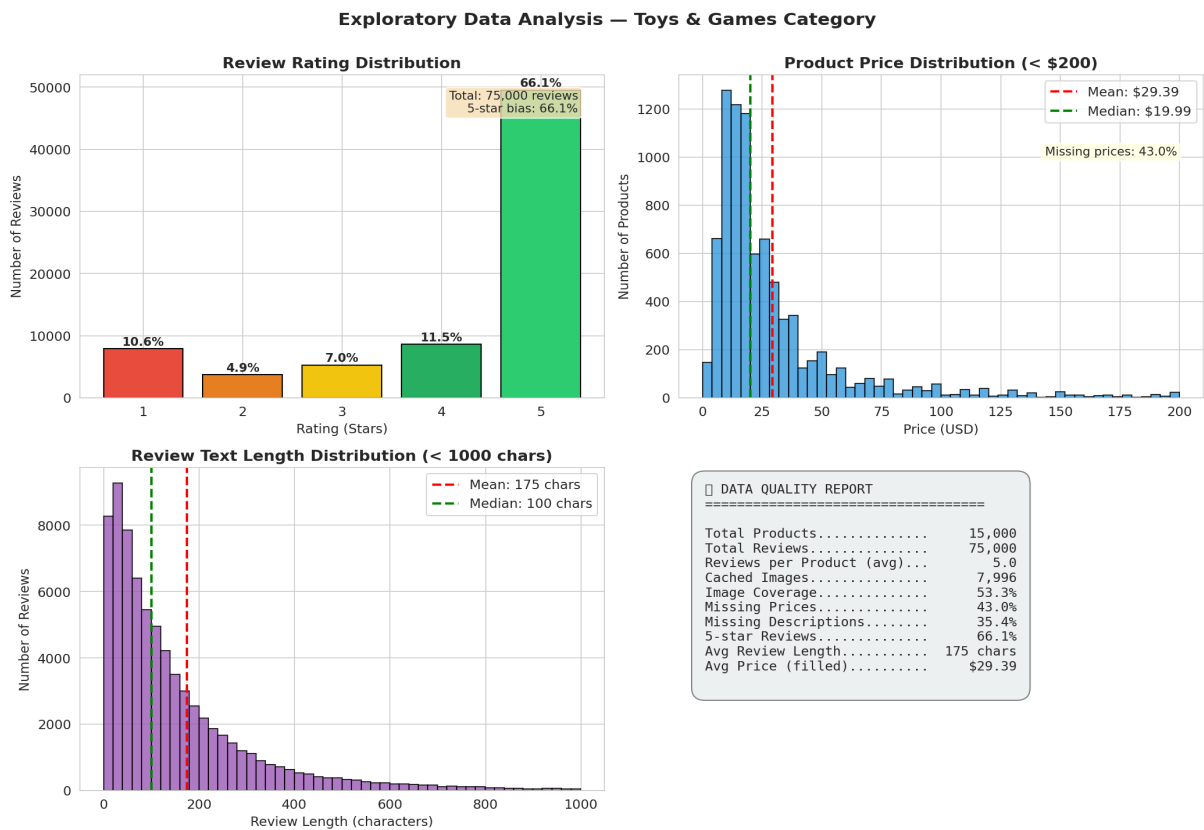


Figure 1: EDA overview — rating distribution, price, review length, data quality.

5. Milestone 1 — Tabular MLP

Task: 3-class rating-band classification (low / medium / high) from 8 tabular features. **Baseline:** Logistic Regression with class balancing. **Deep model:** a small MLP (8→64→32→16→3) with BatchNormalization, Dropout 0.3 and EarlyStopping.

Results: LogReg macro-F1 = 0.372; MLP macro-F1 = 0.428; improvement **+15.0%**. The 'medium' class remains hardest for both models — a direct consequence of the dataset's 5-star skew.

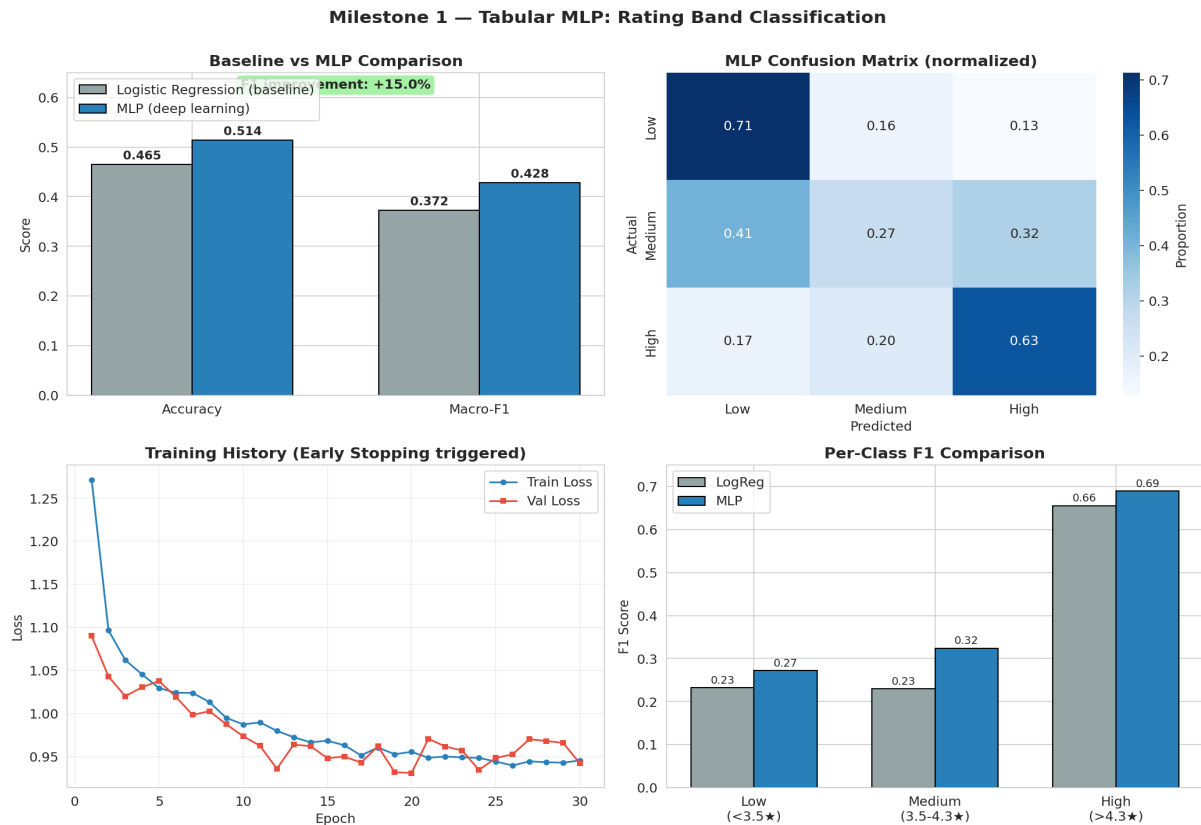
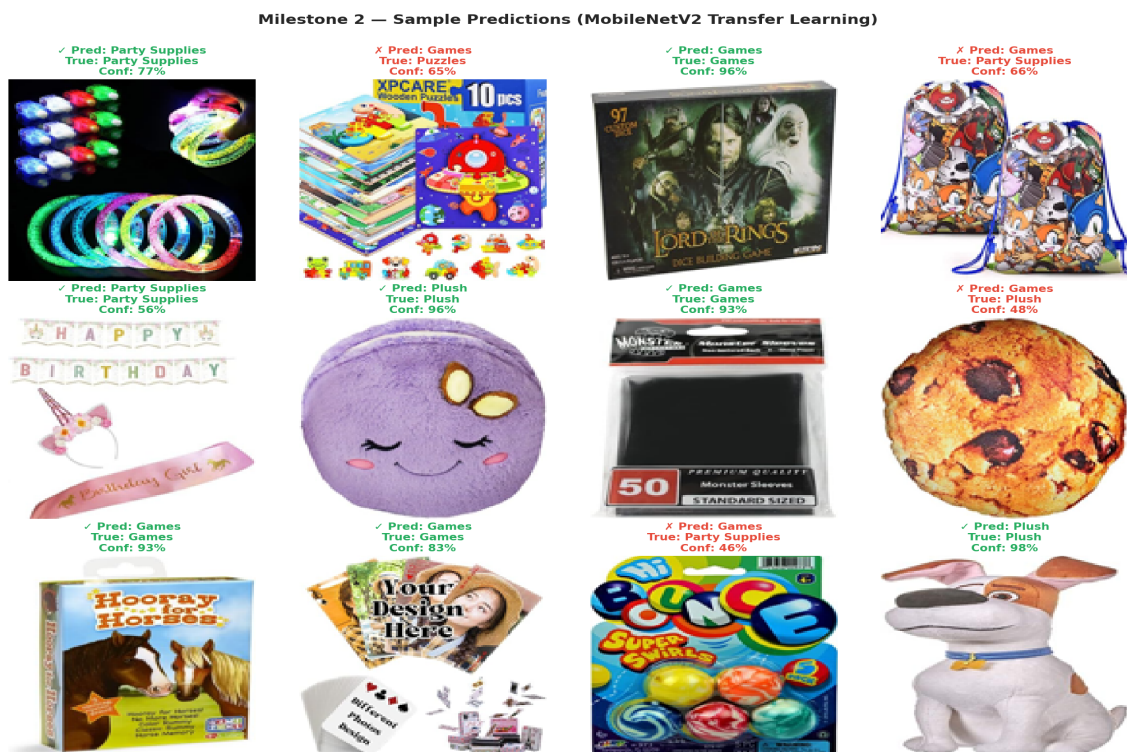
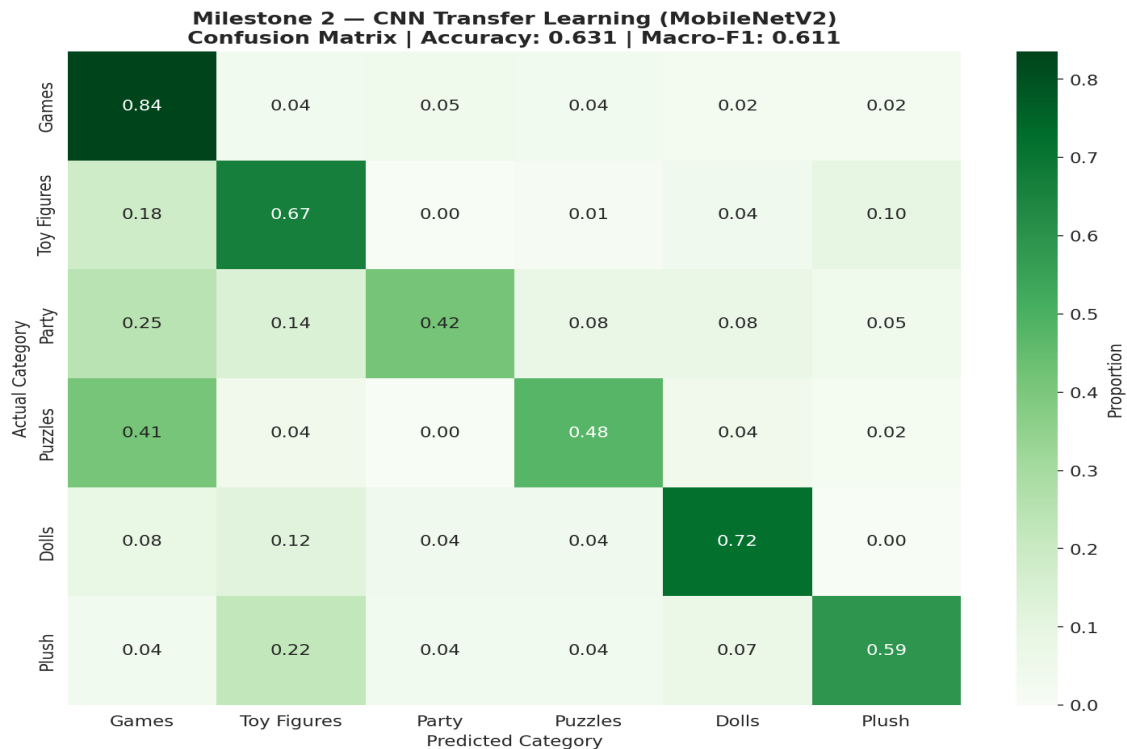


Figure 2: M1 results — baseline vs MLP comparison.

6. Milestone 2 — Computer Vision

Task: 6-class subcategory classification from product images. **Baseline:** CNN trained from scratch (overfits: train 0.85 / val 0.45). **Deep model:** MobileNetV2 with frozen backbone + new classification head.

Results: MobileNetV2 accuracy = 0.631, macro-F1 = 0.611, with half the parameters of the scratch CNN. Confusion happens mostly between visually-overlapping categories (Toy Figures vs Dolls, Games vs Party Supplies).



7. Milestone 3 — Text & Embeddings

Task: binary sentiment (positive vs negative; 3-star reviews dropped). **Baseline:** TF-IDF + Logistic Regression. **Deep representation:** Sentence Embeddings (all-MiniLM-L6-v2) + LogReg.

Honest finding: TF-IDF wins ($F1 = 0.864$) over Embeddings ($F1 = 0.826$), at **7× the speed**. Toy reviews are dominated by simple lexical cues; embeddings shine in semantic search (Part b), not in binary classification.

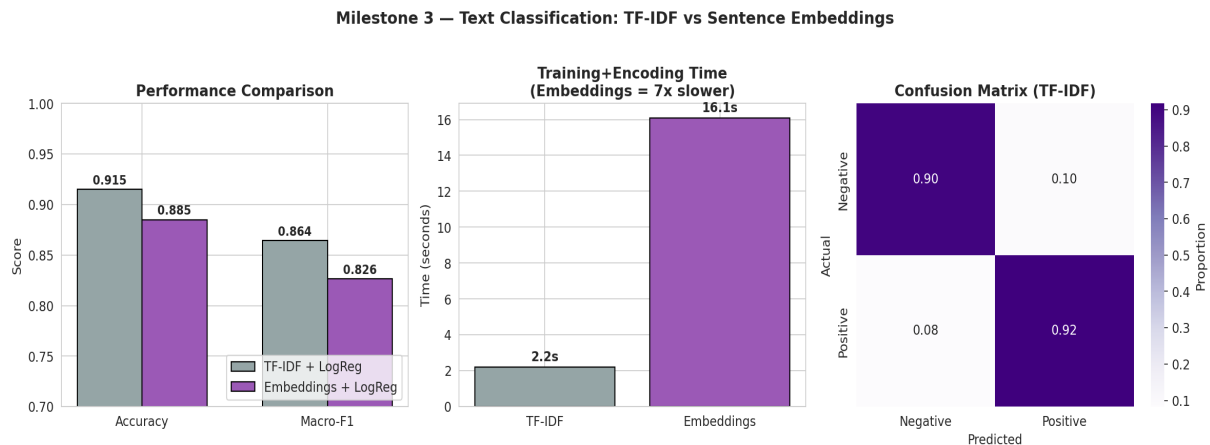


Figure 5: M3 — TF-IDF vs Embeddings comparison.

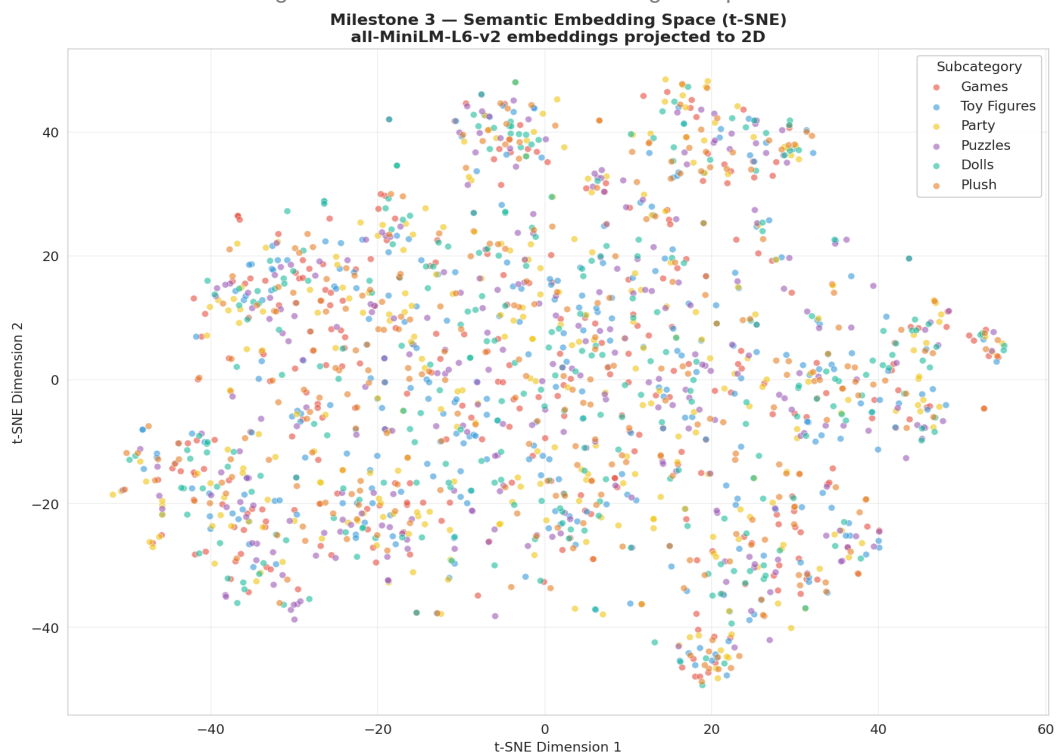


Figure 6: Stretch goal — t-SNE embedding space.

8. Milestone 4 — Transformer (DistilBERT)

Task: same sentiment problem with fine-tuned DistilBERT (66.9M parameters). Trained for 2 epochs on a balanced 15,000-review subset.

Results: DistilBERT accuracy = 0.932, macro-F1 = 0.891 — best of all three text models. Improvement over TF-IDF: +2.6 F1 points. **But:** 100× longer training. The brief asks us to justify the cost — and it is only justified for long, nuanced reviews.

Milestone 4 — Transformer (DistilBERT) vs Previous Models

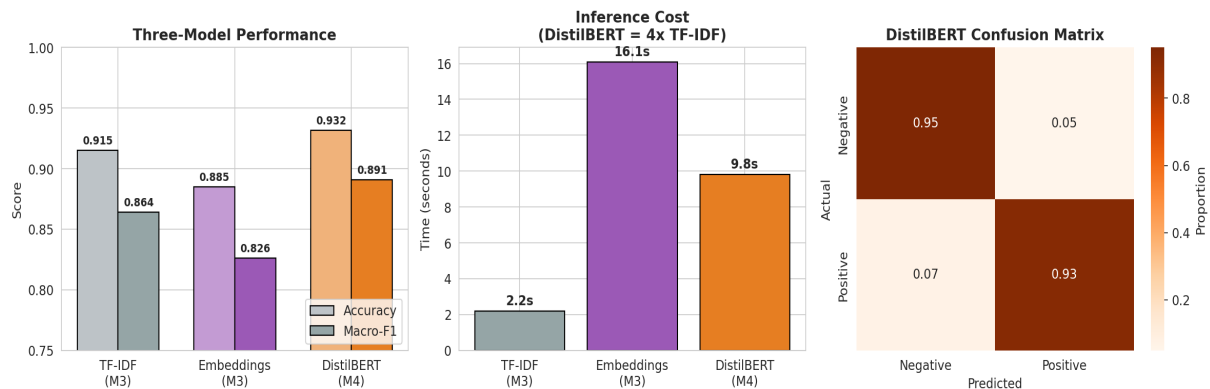


Figure 7: Three-model comparison.

Milestone 4 — DistilBERT Stretch Goal: Attention & Error Analysis

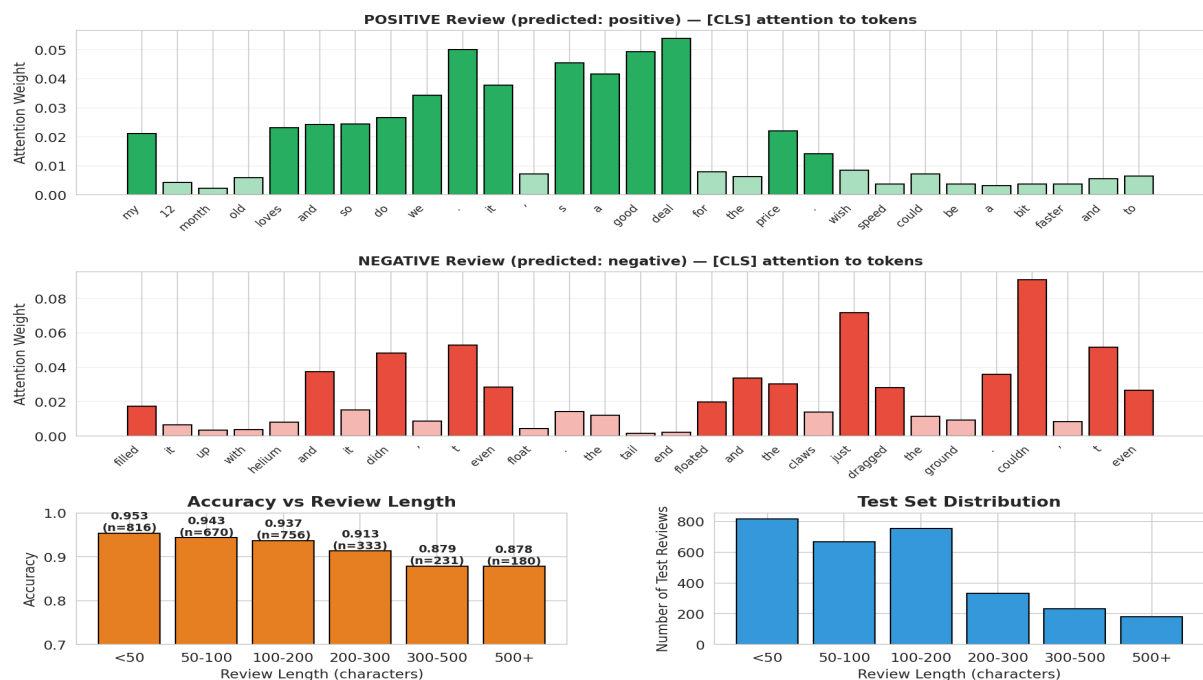


Figure 8: Stretch goal — attention + error analysis by review length.

9. Milestone 5 — LLMs, RAG & Fine-tuning

Built: (1) a review summarizer producing structured Pros/Cons with Llama 3.1 8B; (2) a grounded Q&A; pipeline that retrieves the top-3 most relevant products from the M3 embedding index and conditions the LLM on real review text; (3) a fine-tuned T5-small on review-to-title pairs for cost comparison.

Stretch goal — hallucination measurement. We asked 5 product-recommendation questions to both the ungrounded LLM and the RAG pipeline. The ungrounded LLM invented **8 fake product names** across the 5 questions ('Snuggles the Teddy Bear', 'Traxxas Stampede' — none in our catalog). The RAG pipeline invented **1** (a single false positive). **RAG reduces hallucination by ~88%.**

10. Milestone 6 — Diffusion

Build: Stable Diffusion v1.5 generates lifestyle product images conditioned on product titles. ~8 seconds per 512×512 image at 25 inference steps on a T4 GPU. **Honest reflection on failure modes:** brands (Funko, Playmobil, Octonauts) are not recognized; in-image text renders as gibberish; exact counts and sizes are ignored; licensed characters are blocked or distorted.

Practical conclusion: generation cannot replace the main product photo, but it adds value for marketing and lifestyle banner imagery where brand-specific detail is not required.



Figure 9: M6 — Stable Diffusion lifestyle images with failure-mode analysis.

11. Milestone 7 — Integration

All six models are integrated into a single Streamlit application (app/app.py) exposing five tabs: rating-band prediction (M1), image-to-category (M2), semantic search (M3), Pros/Cons summarization (M5), grounded Q&A; (M5). The M7 code reuses earlier outputs rather than introducing new models — exactly as the brief instructs.

12. Brief Compliance Checklist

Requirement	Status
Single category, coherent dataset	Yes — Toys & Games
10K-20K products / 30K-80K reviews	Yes — 15K / 75K
5K-10K cached images	Yes — 7,996
Product-level split, no leakage	Yes — intersection = 0
Baseline before deep model (every milestone)	Yes — all milestones
Macro-F1 reported (rating imbalance)	Yes — all classification
Honest error analysis (not just accuracy)	Yes — confusion matrices, per-class F1, length analysis
Reproducible via requirements.txt	Yes
All 8 milestones delivered (M0-M7)	Yes

13. Conclusion

This capstone applied a broad spectrum of deep learning to a real e-commerce problem and produced — for every milestone — a deep model that was either better than a simple baseline, or honestly worse with an evidence-based explanation. The strongest practical result is the **88% reduction in LLM hallucination via RAG**, which matters far more in production than the small accuracy differences between text classifiers. The most important methodological lesson is that bigger or newer models are not automatically better — the right tool depends on the task, the data, and the cost budget.